

Quantifying the Value of Improved Minute-Scale Forecasting for Hybrid Power Plants

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Executive Summary

As renewable-energy penetration continues to increase, power systems are becoming increasingly dependent on accurate forecasting to maintain the balance between electricity supply and demand. While significant progress has been achieved in power forecasting, short-term power ramps from wind and solar generation still contain significant forecast uncertainty, leading to large imbalance costs and reserve activation requirements. These challenges are expected to intensify with the penetration of renewable power plants within the energy mix. Hybrid Power Plants (HPPs), combining renewable generation technologies with energy storage behind same grid connection point, offer new opportunities to mitigate renewable variability and make better use of the existing grid infrastructure. Then, improving minute-scale forecasting would enable HPPs to better anticipate renewable fluctuations and improve their operability and profitability. This white paper investigates the value of improved minute-scale ramp forecasting from the perspectives of Hybrid Power Plant operators, energy traders and Balance Responsible Parties (BRPs).

Introduction

The energy transition to mitigate climate change has led to a significant increase of Renewable Energy Sources (RES) in recent years with a progression in Denmark from 15% in 2000 to 87% in 2023 [1]. This shift towards greener energy technologies has mainly been supported through the rise of wind and solar power, however their variability brings new challenges at the system level to correctly balance production [2]. The fluctuations in wind and solar power force Balance Responsible Parties (BRPs) to continuously take actions based on forecasts provided at day-ahead, hours-ahead, and real-time horizons. On the plant energy producer's side, managing volumes accordingly and minimizing their imbalance are driven by the forecast updates.

For wind power forecasts, climatology induced by different topologies is observed to result in different error patterns from days-ahead to hours-ahead forecasts [3]. However, typical Root Mean Square Error for wind power forecasts ranges from 7 to 20% for more than a few hours ahead [3]. Shorter lead times generally lead to more accurate forecasts, with 1-hour-ahead forecasts found to be 35%-55% more accurate than 24-hour-ahead forecasts for 4 European wind farms [4]. For sub-hourly horizons, machine learning (ML) models coupled with site-based LiDAR observations have shown to improve the accuracy of deterministic minute-scale wind speed forecasts by 7-20% over persistence [5]. On the solar side, All-Sky Imagers (ASI) coupled with ML models enable forecasting minute-scale ramp events and improve the accuracy compared to persistence benchmarks by 10-20% depending on the horizon [6, 7, 8]. Therefore, new approaches with LiDAR and ASI are promising to reduce uncertainty in short-term power forecasts and still need to be operationally deployed.

From an energy producer's perspective, Hybrid Power Plants (HPPs) present the advantage of dealing with RES variability by combining two or more independent electricity generation and/or storage technologies providing coordinated electrical power services that are connected at a single point [9]. HPPs are generally subject to ramp constraints. Therefore, HPP ramp forecasting is key to facilitate the operation of Energy Management System (EMS) under grid constraints.

The objective of this white paper is to quantify the economic and operational value of improved minute-scale ramp forecasting for renewable and hybrid power plants. This white paper aims to assess how improved ramp forecasts can improve the profitability of Hybrid Power Plant (HPP) operation. The analysis focuses on the expected impact of a 15% improvement in minute-scale ramp forecasting accuracy.

Case study : DK1

This section quantifies the economic value that improved short-term forecasting brings to HPP operators in DK1. We first characterized ramp events, using CorRES generated raw wind and solar timeseries over twelve locations across DK1. We then examine how EMS dispatch, under stochastic (SO) optimization, converts 30-minute-ahead forecast improvement into measurable balancing-market revenue, and compare the SO and the DO frameworks both in terms of size and consistency of the BM gain across locations.

The analysis is done on each of the locations displayed in the map in Figure 1, where the considered HPP has the following design : Wind 200MW, Solar 100MW, BESS 60MW/180MWh. The single grid connection capacity is set to 200MW. The 300MW combined wind-solar capacity against a 200MW grid connection reflects a 1.5x overplanting ratio, a common practice for hybrid plants to maximize grid connection utilization. This scale is broadly consistent with recent Danish co-location projects, such as the Kvested Energy Park (101MW solar + 50MW/200MWh battery).

Ramp frequency and severity across DK1

Wind and solar production time series, along with their corresponding day-ahead (DA) and 30 minutes-ahead (HA) forecasts, are generated using the CorRES simulation tool [10], which reconstructs meteorologically consistent power time series by combining ERA5 reanalysis data and forecasting model with a stochastic approach that reproduces realistic short-term fluctuations. For this study, CorRES-generated series are used for the 2021 weather year across the twelve DK1 locations considered.

We define a ramp event as an hourly change in aggregated renewable power (wind + solar) output exceeding the 95th percentile of the empirical distribution (VaR_{95}).

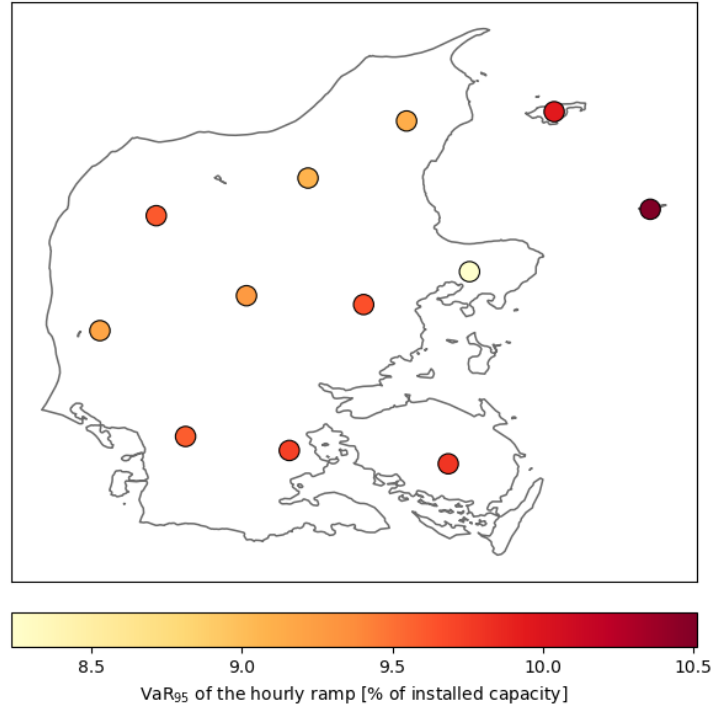


Figure 1: Ramp severity in DK1

The ramp severity differs according to the location across DK1. This heterogeneity indicates that the market risk associated with rapid renewable output changes is not uniformly distributed across the region. Decomposing ramp severity by resource shows that the VaR95 is comparable between wind and solar (13.57% vs. 13.64% of resource-specific capacity on average), with no consistent dominance across locations. This suggests that spatial heterogeneity in overall ramp severity is not explained by one resource dominating over the other, but rather from location-specific weather patterns. It is also seen that the ramp events are more severe on the two islands in the eastern part of DK1. This is consistent with open-water and coastal sites, which typically experience more direct exposure to frontal weather systems compared to inland sites.

Profitability in DK1

The profitability analysis is carried out using HiFiEMS, the high-fidelity energy management system module integrated within the Hydesign framework, whose deterministic and stochastic optimization formulations follow the methodology of [11]. HiFiEMS co-optimizes a day-ahead (DA) market commitment with a subsequent balancing-market (BM) stage, in which regulation capacity is offered around this commitment and settled against realized delivery.

Balancing Market Profitability with the Stochastic Optimizer

Under the stochastic optimizer framework, balancing-market revenue increases from baseline to +15% across all locations (Figure 2), with a mean gain of 2.60%. The baseline scenario represents the case where the 30 minutes-ahead forecasts are exactly the ones generated by CorRES. The +15% scenario applies a uniform 15% reduction in the gap between each 30 minutes-scenario and the corresponding realized measurement, while day-ahead forecasts and market price signals are kept the same across both scenarios. Total revenue is not reported here, as the 30 minutes-ahead forecast improvement only affects the balancing market revenues and imbalance costs.

The spatial variability in balancing-market gain shows a weak negative association with local ramp severity (Pearson $r \approx -0.29$ against the VaR95 of the combined wind+solar ramp), suggesting that locations with lower ramp event severity tend to realize slightly larger gains. However, with only twelve locations, this association is not statistically robust and should be interpreted as suggestive rather than conclusive.

Expressed in EUR/MWh, SO's balancing-market revenue gain under the improved forecast scenario averages 0.153 EUR/MWh across DK1 locations (range: 0.126-0.185 EUR/MWh).

Sensitivity analysis on the benefit of stochastic EMS with enhanced ramp forecast

While Stochastic Optimization (SO) and Deterministic Optimization (DO) achieve statistically similar average balancing-market revenue gains from improved short-term forecasting, SO's key advantage lies in the consistency of this gain across

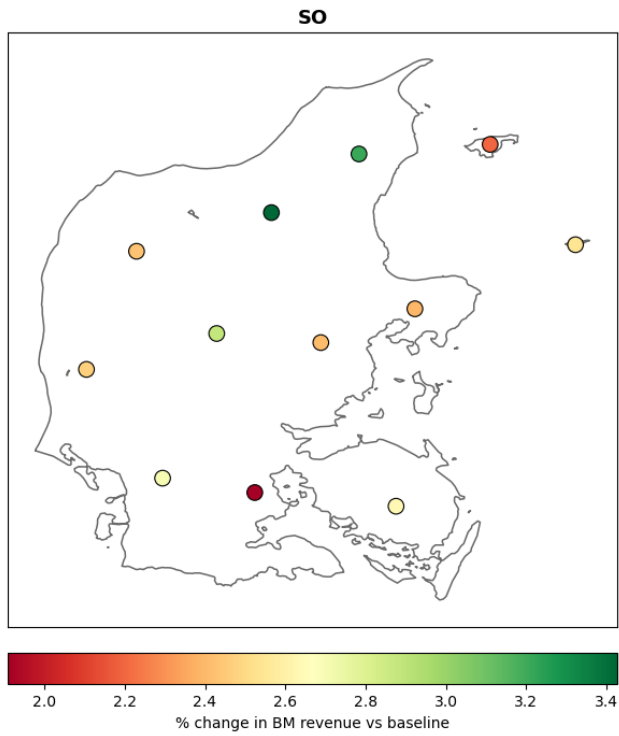


Figure 2: Balancing market relative gain compared to baseline

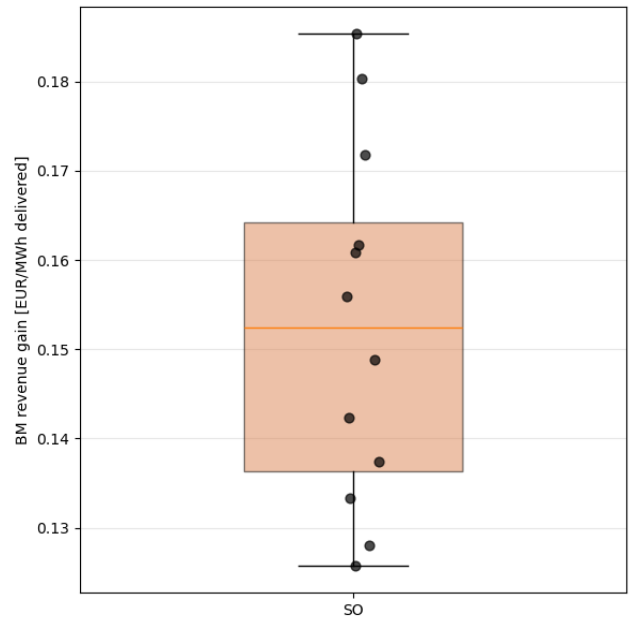


Figure 3: Distribution of the balancing market revenue across all locations, with the Stochastic Optimization

locations. Figure 4 compares the distribution of BM revenue gain across the twelve DK1 locations at +15% forecast improvement, for DO and SO. SO's interquartile range is visibly narrower than DO's, consistent with the standard deviation figures reported above (0.418 vs. 0.735 percentage points). It is also observed that the median value is higher for SO, further supporting SO's more consistently favorable outcome for a typical location.

This lower dispersion is a practically relevant property for portfolio-level deployment: a plant operator managing multiple HPP sites across DK1 benefits from being able to anticipate a consistent return from forecast investment across locations, rather than a highly location-dependent one.

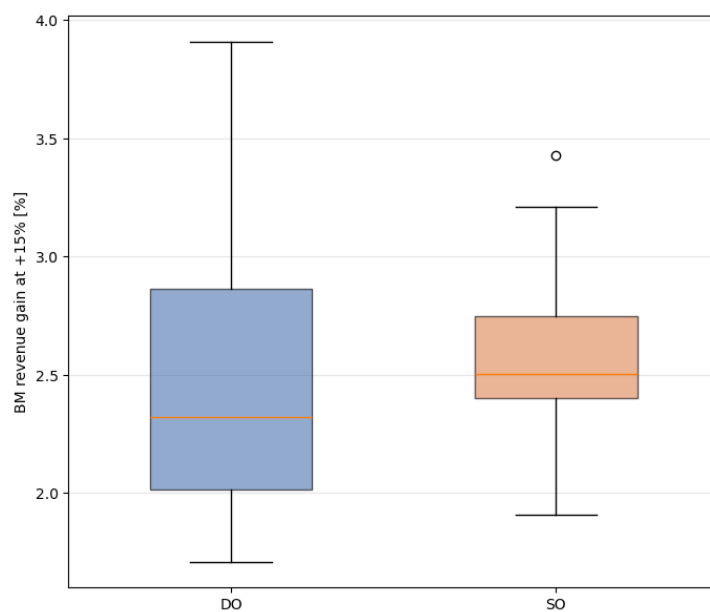


Figure 4: Distribution of the balancing market revenue across all locations, with the Deterministic Optimization (DO) and Stochastic Optimization (SO)

Conclusion

This study quantifies the value of improved 30 minutes-ahead forecasting for hybrid power plants across DK1, using Hi-FiEMS under deterministic (DO) and stochastic (SO) optimization. Both paradigms convert a 15% forecast improvement into **comparable average balancing-market revenue gains (2.5%, or 0.153 EUR/MWh under SO)**. The key difference is reliability, not magnitude: SO's gain is markedly **more consistent** across locations (43% lower standard deviation than DO at 15% forecast improvement), making stochastic, scenario-based dispatch a more predictable return on forecast investment, particularly relevant as the Danish balancing market has moved toward 15-minute settlement.

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